DISTANCE LEARNING PROGRAMME
(Academic Session : 2024 - 2025)

LEADER TEST SERIES / JOINT PACKAGE COURSE

TARGET : PRE-MEDICAL 2025

Test Type : **MAJOR # 03** Test Pattern : **NEET (UG)**

TEST DATE : 16-02-2025

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	2	2	1	3	3	1	3	4	3	2	2	4	3	2	2	2	2	3	3	2	4	1	1	4	2	2	2	1	4
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	4	2	2	1	3	1	1	2	3	3	2	4	4	1	3	2	4	3	2	3	3	2	1	3	2	2	2	2	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	4	3	2	4	2	2	1	4	3	3	1	2	2	1	2	4	2	1	4	3	1	3	2	4	1	3	4	1	1	4
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	3	4	3	3	1	2	3	1	3	4	2	2	2	3	3	3	2	3	2	3	4	1	3	1	1	1	1	3	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	2	4	4	2	4	4	2	1	4	4	2	4	4	4	1	4	1	2	2	1	3	3	4	3	2	3	2	3	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	4	2	1	4	2	2	2	3	3	1	1	1	2	3	2	2	2	1	2	4	3	4	3	4	2	2	2	4	3	3
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	1	2	2	4	3	3	1	1	1	3	4	2	2	4	2	4	4	3	3	3										

HINT – SHEET

SUBJECT : PHYSICS

SECTION - A

1. Ans (2)

Given : $u = 0$, $a_t = 12\ m/s^2$

say, after 't' time, according to question,

$a_{net}\cos 45° = a_c = \frac{v^2}{r}$ ..... (i)

$a_{net}\sin 45° = a_t = 12$ ..... (ii)

Using (i) and (ii),

$v = 6\ m/s$

Now, $v = u + a_t \times t$

$\Rightarrow t = \frac{v-u}{a_t} = \frac{6-0}{12} = \frac{1}{2}s$

2. Ans (2)

Total energy $= -\frac{GMm}{2(R+h)}$, $g = \frac{GM}{R^2}$

$E = -\frac{gR^2m}{2(R+h)}$, $h = R$

$E = -\frac{mgR^2}{2\times 2R} = -\frac{mgR}{4}$

3. Ans (2)

$g_E = g_P - \omega^2 R$

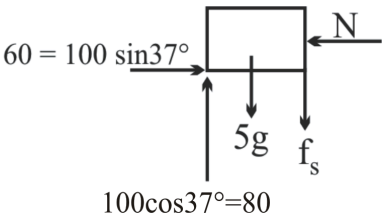
$\Rightarrow \frac{g_p}{2} = g_p - \omega^2 R$

$\Rightarrow \omega = \sqrt{\frac{g_P}{2R}}$

$= \sqrt{\frac{9.8}{2\times 6400\times 10^3}}$

$= 8.75 \times 10^{-4} rad/s$

4. Ans (1)



$f_L = \mu_s N$

$= 0.6 \times 100 \text{ Sin } 37° = 36 \text{ N}$

$F_{req} = 80 - 50 = 30 \text{ N}$

$\therefore f_s = 30 \text{ N}$

5. Ans (3)

If a liquid can rise to a height h, but the tube is of insufficient height h', then the angle of contact increases from θ to θ', given by

$$\frac{h}{\cos\theta} = \frac{h'}{\cos\theta'} \Rightarrow \frac{2}{\cos 0^o} = \frac{1}{\cos\theta'} \Rightarrow \cos\theta' = \frac{1}{2}$$

6. Ans (3)

$6g - T = 6a$ ...(i)

$T = 4a$ ...(ii)

$\therefore 6g = (6 + 4)a$

$$\therefore a = \frac{6g}{10} = \frac{6 \times 10}{10} = 6 \text{ ms}^{-2}$$

7. Ans (1)

$KE_{ring} = KE_{disc}$

$$\Rightarrow \frac{1}{2}mv_R^2(1+1) = \frac{1}{2}mv_d^2\left(1 + \frac{1}{2}\right)$$

$$\Rightarrow \frac{v_R^2}{v_d^2} = \frac{\left(\frac{3}{2}\right)}{2}$$

$\Rightarrow v_d : v_R = \sqrt{4} : \sqrt{3}$

8. Ans (3)

$$\% \text{ error} = \frac{0.001}{0.802} \times 100 = 0.12468 = 0.125\%$$

9. Ans (4)

$$Z = \sqrt{(R)^2 + (X_L - X_C)^2}$$

$R = 10\Omega$, $X_L = \omega L = 2000 \times 5 \times 10^{-3} = 10\Omega$

$$X_C = \frac{1}{\omega C} = \frac{1}{2000 \times 50 \times 10^{-6}} = 10\Omega$$

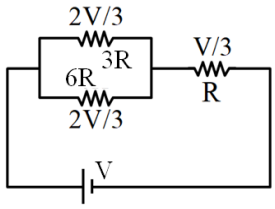
i.e., $Z = 10\ \Omega$

$$\text{Maximum current, } i_0 = \frac{V_0}{Z} = \frac{20}{10} = 2 \text{ A}$$

$$\text{Hence, } i_{rms} = \frac{2}{\sqrt{2}} = 1.4 \text{ A}$$

and $V_{rms} = 4 \times 1.41 = 5.64$ V

10. Ans (3)



$$\frac{4V^2}{9 \times 3R} : \frac{4V^2}{9 \times 6R} : \frac{V^2}{9R}$$

4 : 2 : 3

12. Ans (2)

Here, energy flux, $\phi = 40 \text{ Wcm}^{-2}$, surface area $(A) = 20 \text{ cm}^2$, time $(t) = 10 \text{ min} = 10 \times 60$ s.

Total energy falling on the surface in time t is

$$U = \phi At = 40 \times 20 \times (10 \times 60)\text{J}$$

Momentum of the incident light

$$(P) = \frac{U}{c} = \frac{40 \times 20 \times (10 \times 60)}{3 \times 10^8}$$
$$= 16 \times 10^{-4} \text{ kgs}^{-1}$$

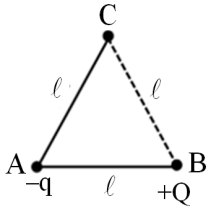
Momentum of the reflected light = 0

$\therefore$ Momentum delivered to the surface

$$= 16 \times 10^{-4} - 0$$

$$= 16 \times 10^{-4} \text{ kg ms}^{-1}$$

13. Ans (4)



$W D = U_f - U_i$

$= \frac{-kqQ}{\ell} - (-)\frac{kQq}{\ell} = 0$

14. Ans (3)

$V_T \propto r^2$

$\frac{V_{T_2}}{V_{T_1}} = \frac{r_2^2}{r_1^2}$

$V_{T_2} = \left(\frac{6}{3}\right)^2 \times 10$

$V_{T_2} = 40\ cm/s$

15. Ans (2)

In first case : $\frac{A}{3} = \frac{B}{4}$

$4A = 3B$.....(i)

In second case

$\frac{A+40}{7} = \frac{B+40}{8}$

$8A + 320 = 7B + 280$ .....(ii)

using equation (i) & (ii),

$6B + 320 = 7B + 280$

$B = 40\Omega$

$A = (3B/4) = 30\Omega$

16. Ans (2)

Time taken by coin to reach the floor is given by :

$h = 1/2\ gt^2 \quad (\because u = 0)$

or $t = \sqrt{\frac{2h}{g}}$

In stationary lift, $t_1 = \sqrt{\frac{2h}{g}}$

In upward moving lift with constant acceleration a,

$g' = g + a$

$\therefore \quad t_2 = \sqrt{\frac{2h}{(g+a)}}$

Clearly, $g' > g$

Thus, $t_2 < t_1$

17. Ans (2)

$t = \frac{2u\sin\theta}{g}$ or $2 = \frac{2u\sin\theta}{g}$

$\therefore u\sin\theta = g$

$\therefore h_m = \frac{u^2\sin^2\theta}{2g} = \frac{g^2}{2g} = \frac{g}{2} = 5m$

18. Ans (2)

The energy of photon $= \frac{hc}{\lambda}$

$\because P = nE$

$P = \frac{nhc}{\lambda}$

$n = \frac{P\lambda}{hc}$

[$\because$ Useful power, $P = 8\%$ of $150\ W = \frac{8}{100} \times 150 = 12W$]

$n = \frac{12 \times 4500 \times 10^{-10}}{6.63 \times 10^{-34} \times 3 \times 10^8}$

$\boxed{n = 27.17 \times 10^{18}}$

19. Ans (3)

$E_2 - E_1 = \frac{hc}{\lambda_2}$ ...(1)

$E_3 - E_1 = \frac{hc}{\lambda_1}$ ...(2)

$E_3 - E_2 = \frac{hc}{\lambda_3}$ ...(3)

Add equation (1) & (3)

$E_2 - E_1 + E_3 - E_2 = \frac{hc}{\lambda_2} + \frac{hc}{\lambda_3}$

$E_3 - E_1 = \frac{hc}{\lambda_2} + \frac{hc}{\lambda_3}$

$\frac{hc}{\lambda_1} = \frac{hc}{\lambda_2} + \frac{hc}{\lambda_3}$

$\boxed{\frac{1}{\lambda_1} = \frac{1}{\lambda_2} + \frac{1}{\lambda_3}}$

20. Ans (3)

Energy will be released when stability increases. This will happen when binding energy per nucleon increases.

21. Ans (2)

No. of α particles = 8

No. of β^- particles = 4

No. of β^+ particles = 2

So, for daughter nucleus,

$z = 92 - (8 \times 2) + (4 \times 1) - (2 \times 1)$

(for α) (for β^-) (for β^+)

$z = 92 - 16 + 4 - 2$

$\boxed{z = 78}$

22. Ans (4)

$\because m = \dfrac{f}{f-u} = \dfrac{h_i}{h_0}$

$\Rightarrow \dfrac{20}{20+20} = \dfrac{h_i}{2 \times 10^{-3}}$

$\Rightarrow h_i = 10^{-3}$ m = 1 mm

24. Ans (1)

$I_L = \dfrac{V_Z}{R_L} = \dfrac{5}{2}$mA = 2.5mA

25. Ans (4)

$Y = \overline{(AB)}.(A+B) = (\bar{A}+\bar{B})(A+B)$

$= \bar{A}.A + \bar{A}B + \bar{B}A + \bar{B}.B$

$= \bar{A}B + A\bar{B}$

26. Ans (2)

$TV^{\gamma-1}$ = constant

] $x = \gamma - 1$

On comparing with TV^x = constant,

$x = \gamma - 1 = \dfrac{5}{3} - 1 = \dfrac{2}{3}$

27. Ans (2)

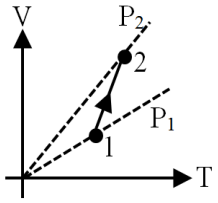
$\dfrac{1}{2}\left(\dfrac{1}{2}mV^2\right) = mS\Delta T$

$\Delta T = 150°C$

28. Ans (2)

$\tan\theta$ = slope = $\dfrac{V}{T} = \dfrac{nR}{P}$

Slope $\propto \dfrac{1}{P}$



29. Ans (1)

$M = IA$

$M = [A]\,[L^2]$

$= [M^0L^2T^0A^1]$

30. Ans (4)

Ratio of freq = 375 : 625 : 875

= 3 : 5 : 7 (odd ratio)

so organ pipe is closed

fundamental freq = $\dfrac{(625-375)}{2}$ = 125 Hz

31. Ans (3)

$T \downarrow \longrightarrow$ freq $\downarrow$

If original freq of B were less than that of A, further decrease in ν_B should have resulted in an increase in beats. But beat frequency is found to decrease on reducing tension in string B,

so originally, $\nu_B > \nu_A \Rightarrow \nu_B = 255 + 5 = 260$ Hz

32. Ans (4)

$\beta \propto \lambda$

$\beta' = \dfrac{\lambda'}{\lambda}\beta = \dfrac{6400}{4800} \times 0.03$mm

= 0.04 mm

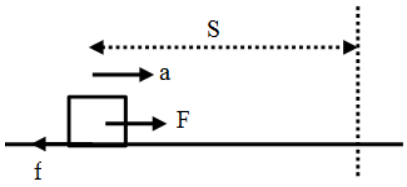
33. Ans (2)

$\alpha = \dfrac{\lambda}{a} = \dfrac{6328 \times 10^{-10}}{0.2 \times 10^{-3}}$rad

$= 0.18^o$

$\left(\because 1\text{ rad} = \dfrac{180^o}{\pi}\right)$

34. Ans (2)



$f = (0.2)(4)(10) = 8N$

$a = \frac{F - f}{M} = \frac{18 - 8}{4} = \frac{10}{4} = 2.5\ m/s^2$

$S = \frac{1}{2}at^2 = \frac{1}{2} \times 2.5 \times 100 = 125\ m$

$W = \vec{F}.\vec{S} = 18 \times 125 \times 1 = 2250$ Joule

35. Ans (1)

$BI\ell = mg$

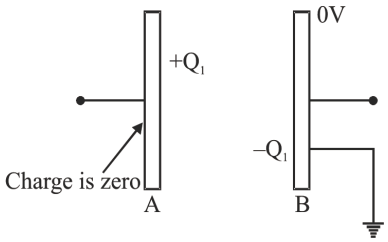
$B = \frac{mg}{I\ell} = \left(\frac{m}{\ell}\right)\frac{g}{I} = 45 \times 10^{-3} \times \frac{10}{30}$

$= 15 \times 10^{-3}$ T

SECTION - B

36. Ans (3)

When we earth plate B, whole of its charge will go to earth. Now charge of plate A will induce charge $-Q_1$ and $+Q_1$ on inner side and outer side of plate B, respectively. Again $+Q_1$ charge of plate B will go to earth. $-Q_1$ charge will not go to earth. It will be bounded to $+Q_1$ charge of plate A.



So, it is clear that $Q_1 + Q_2$ charge goes from 2nd plate to earth.

Charge on capacitor is Q_1 and hence its potential is $\frac{Q_1}{C}$.

37. Ans (1)

$4(H - 4) = 6(H - 6)$

or $2H = 36 - 16 = 20$ or $H = 10$ cm

38. Ans (1)

Assertion is true but reason is false.

As no torque is applied, angular momentum remains constant

$I_1\omega_1 = I_2\omega_2$

$\therefore \quad \frac{\omega_2}{\omega_1} = \frac{I_1}{I_2}$

Now, $\frac{E_2}{E_1} = \frac{(1/2)I_2\omega_2^2}{(1/2)I_1\omega_1^2} = \frac{I_2}{I_1}\left(\frac{I_1}{I_2}\right)^2 = \frac{I_1}{I_2}$

Clearly, if moment of inertia decreases, KE of rotation increases and vice-versa.

39. Ans (2)

$\frac{Q}{A\varepsilon_0K_1} - \frac{Q}{A\varepsilon_0K_2} = x$

and $\frac{Q}{A\varepsilon_0K_2} - \frac{Q}{A\varepsilon_0K_3} = y$

Put in $2x = y$

We get $\frac{3}{K_2} = \frac{2}{K_1} + \frac{1}{K_3}$

40. Ans (3)

$\Delta KE = \frac{1}{2}\frac{m_1m_2}{m_1 + m_2}(1 - e^2)(u_1 - u_2)^2$

$\Rightarrow e = \frac{1}{\sqrt{2}}$

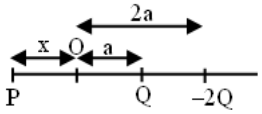
41. Ans (3)

$e = VB\ell = 20 \times 2 \times 10^{-2} \times 20 \times 10^{-2}$

$= 0.08$ V

$I = 0$

42. Ans (2)



Let, At point P, E = 0, then

$$\frac{KQ}{(x+a)^2} = \frac{K.2Q}{(x+2a)^2}$$

$$\Rightarrow \sqrt{2}(x+a) = x+2a$$

$$\Rightarrow x = \frac{2a-\sqrt{2}a}{\sqrt{2}-1} = \sqrt{2}a$$

So, resultant electric field is zero at point 'P" which has position as,

$x = -\sqrt{2}a\,\hat{i}$

43. Ans (4)

$$\lambda_p = \frac{h}{\sqrt{2m_pq_pV_p}} \quad ...(1)$$

$$\lambda_{Be} = \frac{h}{\sqrt{2m_{Be}q_{Be}V_{Be}}} \quad ...(2)$$

$\lambda_P = \lambda_{Be}$

$$\frac{h}{\sqrt{2m_pq_pV_p}} = \frac{h}{\sqrt{2m_{Be}q_{Be}V_{Be}}}$$

$m_pq_pV_p = m_{Be}\,q_{Be}\,V_{Be}$

$m_pq_pV_p = 8m_p \times 2q_p \times V_{Be}$

$[\because m_{Be} = 8m_p,\ q_{Be^{+2}} = 2q_p]$

$16V_{Be} = V_p$

$$V_{Be} = \frac{100}{16} = 6.25 \text{ volt.}$$

44. Ans (4)

Intensity reduced therefore saturation current reduced.

Frequency increased, therefore stopping potential increased.

45. Ans (1)

TE = 300 J

$$TE = U_0 + \frac{1}{2}KA^2 = 300 \text{ J}$$

U_0 is PE at M.P. = 100 J

$$\frac{1}{2}KA^2 = 300 - 100 = 200 \text{ J}$$

$$KE = \frac{1}{2}K\left(A^2 - x^2\right)$$

$$KE \text{ at } x = \frac{A}{\sqrt{2}}$$

$$KE = \frac{1}{2}k\left(A^2 - \frac{A^2}{2}\right) = \frac{1}{4}KA^2$$

$$KE = \frac{1}{4} \times 400 = 100 \text{ J}$$

46. Ans (3)

$$\mu = \frac{\sin\left(\frac{\delta_{min}+A}{2}\right)}{\sin A/2} \Rightarrow \frac{3}{2} = \frac{\sin A}{\sin A/2}$$

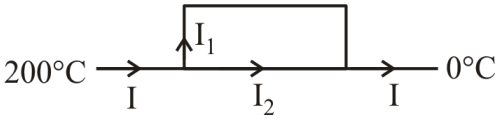
$$\Rightarrow \frac{3}{2} = \frac{2\sin A/2\cos A/2}{\sin A/2}$$

$$\Rightarrow \cos\frac{A}{2} = \frac{3}{4} = 0.75$$

$$\frac{A}{2} = 41°$$

$$A = 82°$$

47. Ans (2)



$$I_1 = \frac{R_2 I}{R_1 + R_2} \qquad [R \propto \ell]$$

$$\frac{60 \times 150}{120+60} = \frac{60 \times 150}{180} = 50\text{J}$$

48. Ans (4)

$$\vec{S}_1 = 30\left(\frac{3\hat{i}+4\hat{j}}{5}\right) = 18\hat{i} + 24\hat{j}$$

$\vec{S}_2 = A_x\hat{i} - A_y\hat{j}$

$\vec{S}_{net} = \vec{S}_1 + \vec{S}_2$

$= (18 + A_x)\hat{i} + (24 - A_y)\hat{j}$

S_{net} only along x-axis, so

$24 - Ay = 0$

$A_y = 24$ m

Now, $\vec{S}_1 . \vec{S}_2 = 0$

$18\ A_x = (24)(24)$

$A_x = 32$

$d = |\vec{S}_2|$

$= |32\hat{i} - 24\hat{j}|$

$= 40$m

49. Ans (3)

$$\frac{B_{in}}{B_{out}} = \frac{\mu_0 Ir'}{2\pi R^2} \times \frac{2\pi r}{\mu_0 I} = \frac{\mu_0 I(a/3)}{2\pi a^2} \times \frac{2\pi(2a)}{\mu_0 I}$$

50. Ans (2)

only (b) is incorrect

SUBJECT : CHEMISTRY

SECTION - A

51. Ans (3)

Highest rate of increment in successive IE is between IE_3 and IE_4 hence, M has three valence e^- and its possible oxidation state is +3.

52. Ans (3)

He ; Nobel gas have zero EN

53. Ans (2)

NCERT XI Pg. # 113 (Part-I)

54. Ans (1)

$2PCl_5(s) = [PCl_4]^+ [PCl_6]^-$ (theoretical factual)

55. Ans (3)

NCERT XIth Pg.#108 Para-1 (Part-I)

56. Ans (2)

Stability $\propto$ Bond order

57. Ans (2)

sp^3 – tetrahedral, dsp^2 - square planar

sp^3d – trigonal bipyramidal, d^2sp^3-octahedral

58. Ans (2)

Trans isomer

59. Ans (2)

In p-block metals o.s. differs by 2.

eg. $\longrightarrow Pb^{+2}, Pb^{+4}$

60. Ans (1)

Generally basicity decreases as we move from left to right to period so the most basic lanthanoid here is $Pr(OH)_3$.

61. Ans (4)

$Na_2Cr_2O_7$ is hygroscopic so don't use is volumetric analysis.

62. Ans (3)

% of N in $(NH_2)_2CO = \frac{28}{60} \times 100 = 46.7\%$

% of N in $NH_3 = \frac{14}{17} \times 100 = 82.3\%$

% of N in $NH_4NO_3 = \frac{28}{80} \times 100 = 35\%$

% of N in $HNC(NH_2)_2 = \frac{42}{59} \times 100 = 71.1\%$

Lower the percentage of N in the fertilizer, lower is its price, hence 50 kg bag of NH_4NO_3 is least expensive.

64. Ans (4)

The unit of K_c is mol L^{-1} which is only possible for the reaction

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]}$

Unit $\longrightarrow$ mol L^{-1}

65. Ans (2)

Initially equal moles of A and B are taken, while rate of disappearance of A is twice of B hence at any time during the reaction or at equilibrium [A] < [B].

66. Ans (2)

On increasing temperature, degree of ionisation increases, $[H^+]$ & $[OH^-]$ both increases hence pH$\downarrow$

67. Ans (1)

$A_{(g)} + B_{(g)} \rightleftharpoons 2C_{(g)}$

$k_C = \frac{[C]^2}{[A][B]}$

$4 = \frac{[C]^2}{0.1 \times 0.4}$

$[C]^2 = 0.16$

$[C] = 0.4$ M

68. Ans (4)

Vapour pressure is independent of surface area.

70. Ans (3)

In first order reaction

$$t = \frac{2.303}{k}\log\frac{[A]_0}{[A]_t}$$

$$10 = \frac{2.303}{k}\log\frac{100}{50} \quad ...(i)$$

$$t_{1/4} = \frac{2.303}{k}\log\frac{100}{75} \quad ...(ii)$$

Divide (i) and (ii)

$t_{1/4}$ or $t_{25\%} = 4$ days

72. Ans (2)

$$\frac{1}{2}Hg_2Cl_2(s) + e^- \rightarrow Hg(\ell) + Cl^-(aq)$$

$n = 1, Q = [Cl^-]$

$$E = E^o - \frac{0.059}{1}\log Q$$

$$= E^o - \frac{0.059}{1}\log[Cl^-]$$

$[Cl]\downarrow, E\uparrow$

79. Ans (4)

Ketone does not reduce Fehling solution

80. Ans (3)

Structural isomeric alcohol of $C_4H_{10}O$ is

$CH_3CH_2CH_2CH_2-OH$

$CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-OH$

$CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-OH$

$CH_3-\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}-OH$

81. Ans (1)

Reactivity of halide $\propto$ Stability of carbocation

85. Ans (1)

$CH_3(CH_2)_3-OH > C_2H_5-O-C_2H_5 > CH_3CH_2CH_2CH_3 > CH_3-CH_3$

H – bonding; Dipole – Dipole Attraction; B.P $\propto$ Mole. mass

SECTION - B

86. Ans (3)

NH_3 forms H-bond with water.

87. Ans (4)

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88. Ans (1)

$$\text{Thermal stability of oxy salt} \propto \frac{1}{\text{Polarisation}}$$

89. Ans (1)

Brown ring test

$$NO_3^- + Fe^{+2} + 4H^+ \rightarrow Fe^{+3} + NO + 2H_2O$$

$$[Fe(H_2O)_6]SO_4 + NO \rightarrow [Fe(H_2O)_5NO]SO_4$$

Brown ring complex

90. Ans (4)

(A) For a given metal and ligand the stability is greater when the charge on the metal ion is greater; as oxidation state increases size of cation decreases.

(B) F^- forms strongest complex. Smaller the size of the anion, greater the charge density on the atom and hence greater the sability of complex.

(C) Smaller the size of the metal ion, greater is the stability of the complex.

91. Ans (4)

$$CuSO_4 + KI \rightarrow Cu_2I_2 + I_2 + K_2SO_4$$

I_2 + Starch $\rightarrow$ Blue colour

92. Ans (3)

$$\Delta H^o_{comb}(H_2) = \Delta H^o_f(H_2O)$$

$$\text{Calorific value} = \frac{\Delta H^o_{comb}(H_2)}{Mw_{H_2}}$$

$$-143 = \frac{\Delta H^o_f(H_2O)}{2}$$

$$\Delta H^o_f(H_2O) = -286 \text{ kJ mol}^{-1}$$

93. Ans (4)

$\Delta G = \Delta H - T\Delta S$

if ΔH is +ve and ΔS is –ve then ΔG will be positive at all temperatures.

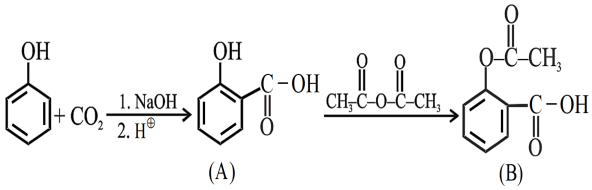
94. Ans (3)

Smaller the valve of K_a, weaker will be the acid and stronger will be its conjugate base.

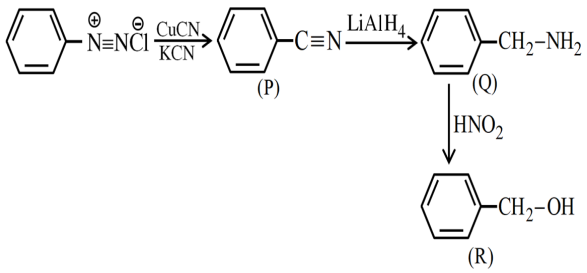
95. Ans (3)

Oxidation number of S in H_2SO_4 is +6 which is maximum. hence it can not further oxidise and can not behave as reducing agent.

98. Ans (3)



100. Ans (3)



SUBJECT : BOTANY

SECTION - A

101. Ans (4)

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102. Ans (2)

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103. Ans (2)

NCERT-XII, Pg. # 55

104. Ans (2)

NCERT-XII Pg#61

105. Ans (3)

NCERT XII, Pg # 67,70

106. Ans (3)

NCERT XII Pg. # 72

107. Ans (3)

NCERT XII Pg.# 93

108. Ans (2)

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109. Ans (3)

NCERT XII Pg. # 90

110. Ans (2)

NCERT-XII, Pg # 103

111. Ans (3)

NCERT XII Pg. No. # 207

112. Ans (4)

NCERT XII Pg. # 210

113. Ans (1)

NCERT Pg. # 219

114. Ans (3)

NCERT Pg. # 223,224

115. Ans (1)

NCERT XII Pg. # 197,198

116. Ans (1)

NCERT Pg # 194

117. Ans (1)

NCERT Pg. # 201

118. Ans (1)

NCERT XI Pg. # 24

119. Ans (3)

NCERT Pg. # 17, 18

120. Ans (2)

NCERT Pg # 13

121. Ans (4)

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122. Ans (2)

Ephedra is vessel containing gymnosperm

123. Ans (4)

NCERT XI Pg. No. # 75

124. Ans (4)

NCERT-XI, Pg. # 77

125. Ans (2)

NCERT-XI, Pg. # 72

126. Ans (4)

NCERT Pg. No. # 67

127. Ans (4)
NCERT-XI, Pg # 67

128. Ans (2)
NCERT XI Pg # 60

129. Ans (1)
NCERT Pg. # 177

130. Ans (4)
NCERT Pg. # 175, 176

131. Ans (4)
NCERT Pg. # 142, 143

132. Ans (2)
NCERT Pg. # 150

133. Ans (4)
NCERT Pg. # 139

134. Ans (4)
NCERT XI Pg. # 159

135. Ans (4)
NCERT XI Pg.# 18,19,20

SECTION - B

136. Ans (1)
NCERT Pg. # 94

137. Ans (4)
NCERT XII, Page # 210

139. Ans (2)
NCERT (XII) Pg. # 62

142. Ans (3)
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143. Ans (3)
NCERT XI Pg. No. # 76

144. Ans (4)
NCERT, 58, 59, 61

145. Ans (3)
NCERT-XII, Pg. # 65

146. Ans (2)
NCERT-XI, Pg # 118

147. Ans (3)
NCERT Pg. # 145

148. Ans (2)
NCERT Pg. # 157

149. Ans (3)
NCERT XI, Pg. # 159

150. Ans (4)
NCERT-XI, Pg # 38

SUBJECT : ZOOLOGY

SECTION - A

151. Ans (4)
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152. Ans (2)
NCERT Pg. # 47

157. Ans (2)
NCERT Pg. # 200, 15.3.2

158. Ans (3)
NCERT Pg. # 194, 15.1.2

160. Ans (1)
NCERT-XII, Pg. # 185-187

164. Ans (3)
NCERT-XI, Pg. # 333, 335

169. Ans (2)
NCERT, Pg # 37

180. Ans (3)
NCERT XIth Pg.# 163, 164

184. Ans (4)
NCERT XII Pg # 183,184

SECTION - B

186. Ans (3)

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187. Ans (1)

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191. Ans (4)

NCERT Pg. # 203, 15.6

192. Ans (2)

NCERT Pg. # 187

195. Ans (2)

NCERT Pg. No. # 310, 311

196. Ans (4)

NCERT P.No. 32

197. Ans (4)

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